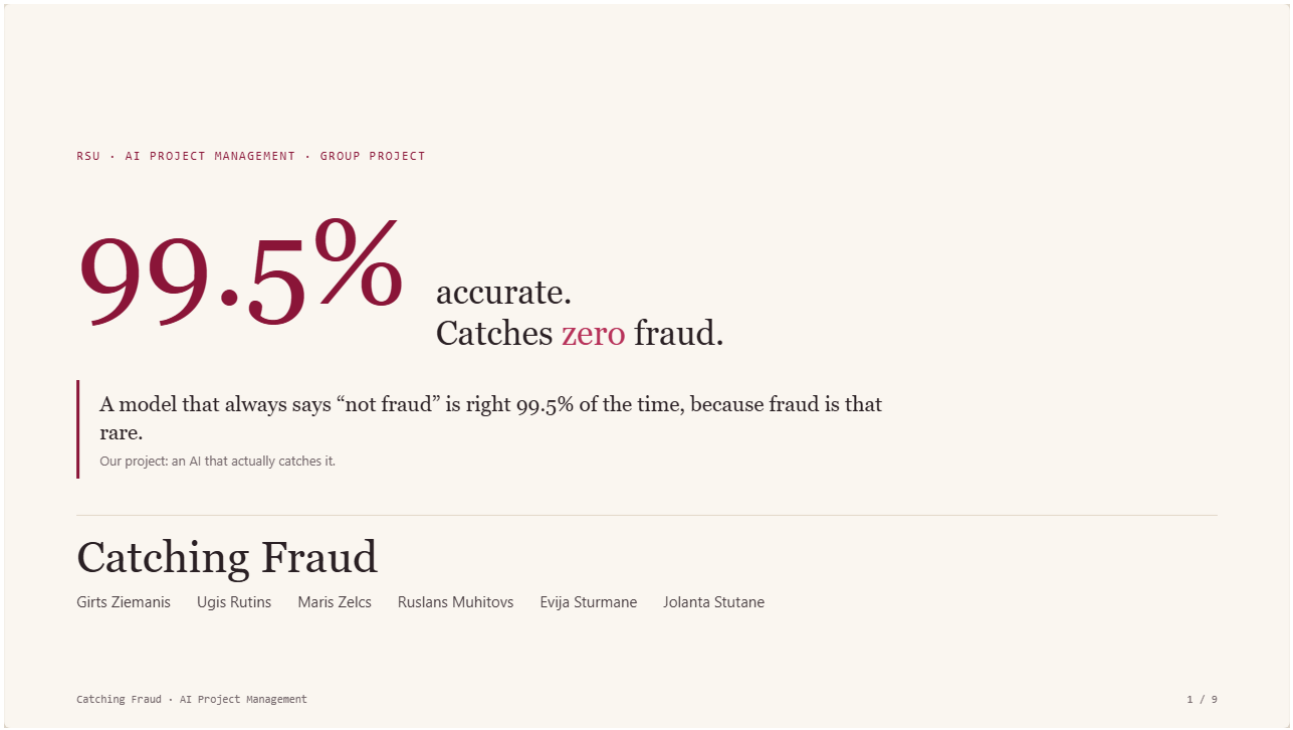
SLIDE 1 / 9 · RSU · AI PROJECT MANAGEMENT · GROUP PROJECT

Hook

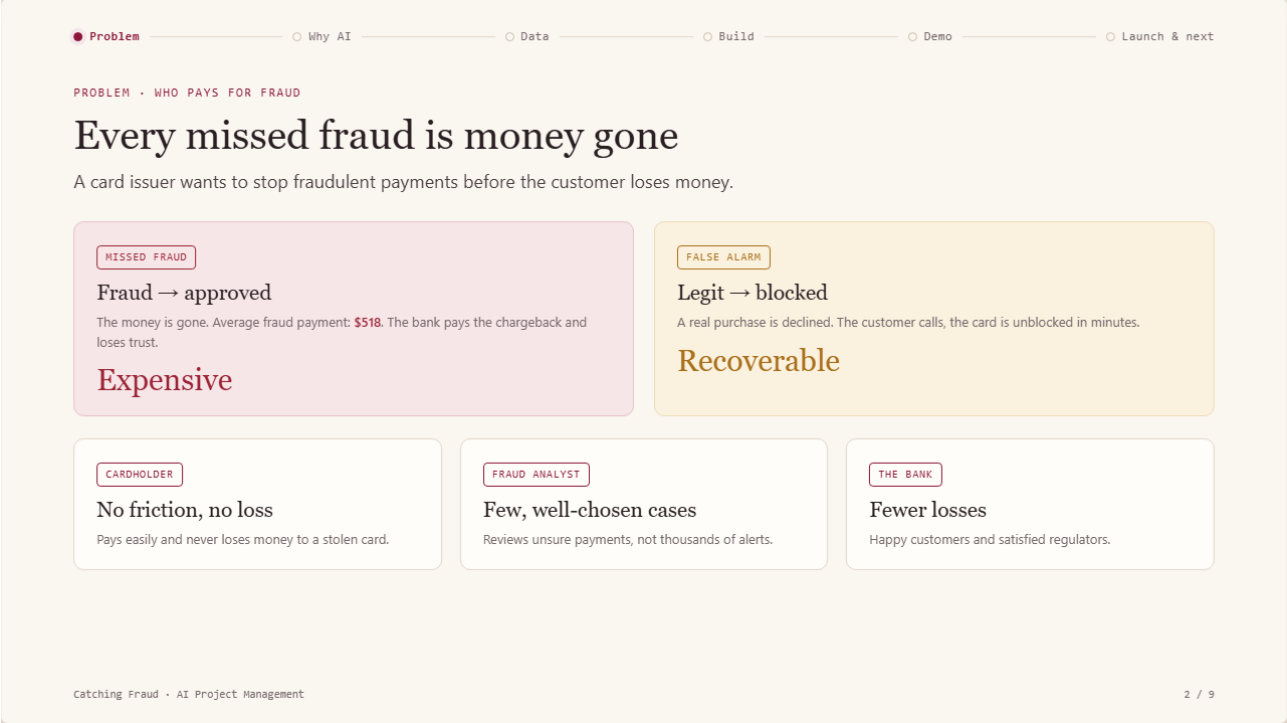


SPEAKER NOTES

- (~20 s) "Imagine a fraud detector that is 99.5% accurate. Sounds great? It catches zero fraud."
- Only 0.52% of card payments are fraud, so a model that always says "not fraud" is right 99.5% of the time and useless.
- "We are team Catching Fraud, and in the next minutes we show you an AI that does catch it: where its data comes from, why it needs AI, and how we'd launch it."

SLIDE 2 / 9 · PROBLEM · WHO PAYS FOR FRAUD

The problem

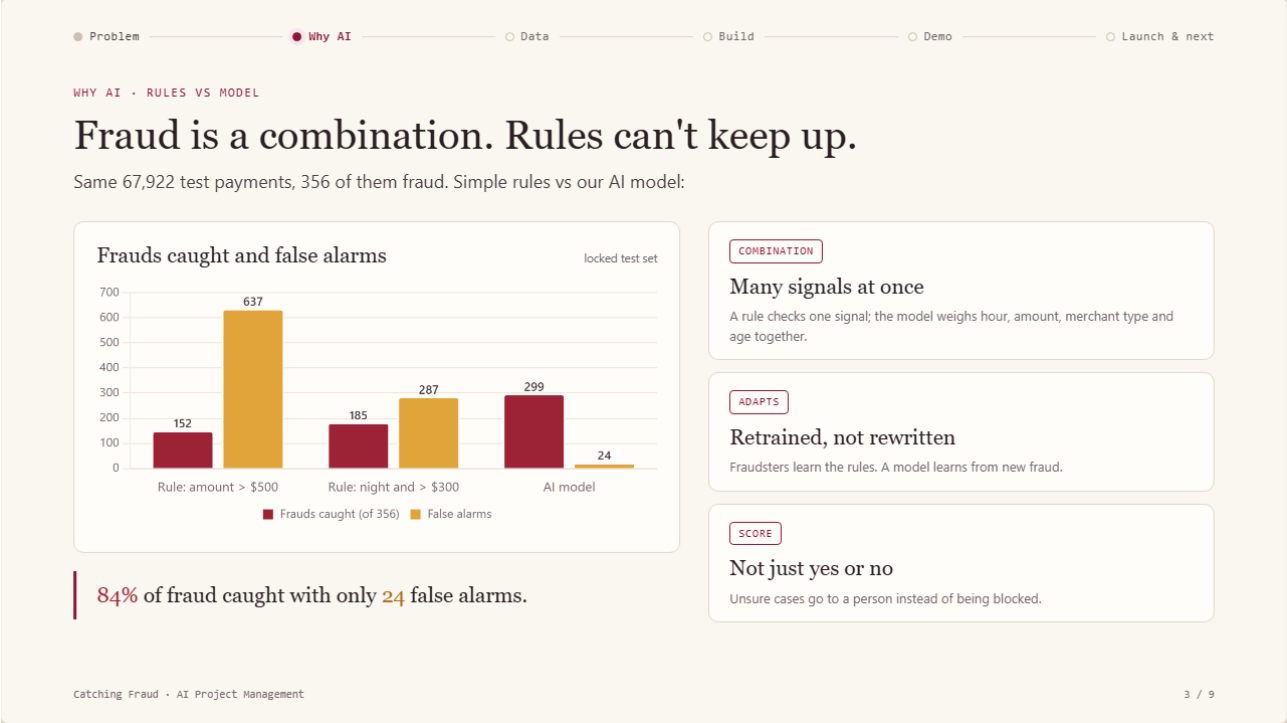


SPEAKER NOTES

- (~40 s) Two possible mistakes, very different prices. A missed fraud is real money lost (average $518 in our data) plus a chargeback and lost trust. A false alarm is an annoyed customer and a phone call.
- So our goal is: catch as much fraud as possible, while keeping false alarms low. "When in doubt, stop the transaction."
- Three users: the cardholder, the fraud analyst who reviews cases, and the bank that pays for losses.

SLIDE 3 / 9 · WHY AI · RULES VS MODEL

Why AI

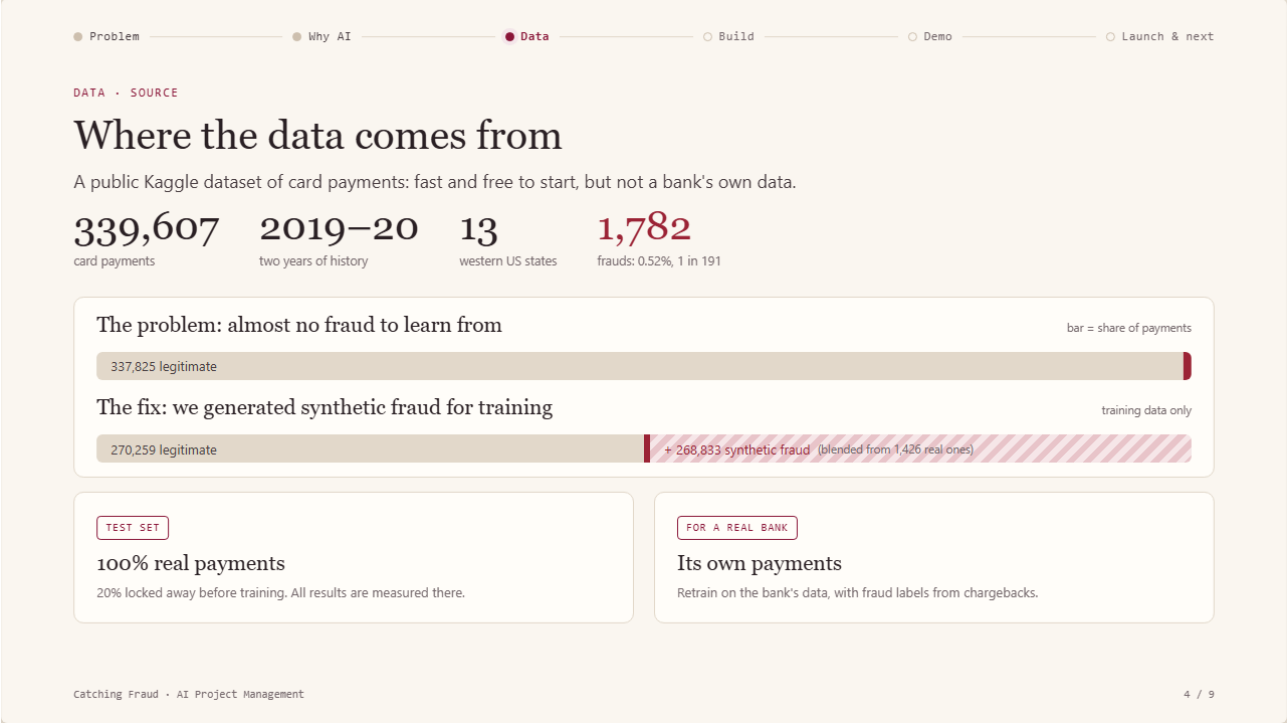


SPEAKER NOTES

- (~45 s) Why do we need AI at all? We tested the obvious rules on exactly the same test payments as the model.
- "Block every payment over $500" catches 152 of 356 frauds (43%) and blocks 637 innocent customers. "Block night payments over $300" catches 185 (52%) with 287 false alarms.
- Our AI model catches 299 (84%) with only 24 false alarms. It combines many weak signals that a rule looks at one by one.
- And fraud changes: rules must be rewritten by hand, a model is retrained. It also gives a score, so unsure cases can go to a person.

SLIDE 4 / 9 · DATA · SOURCE

Where the data comes from

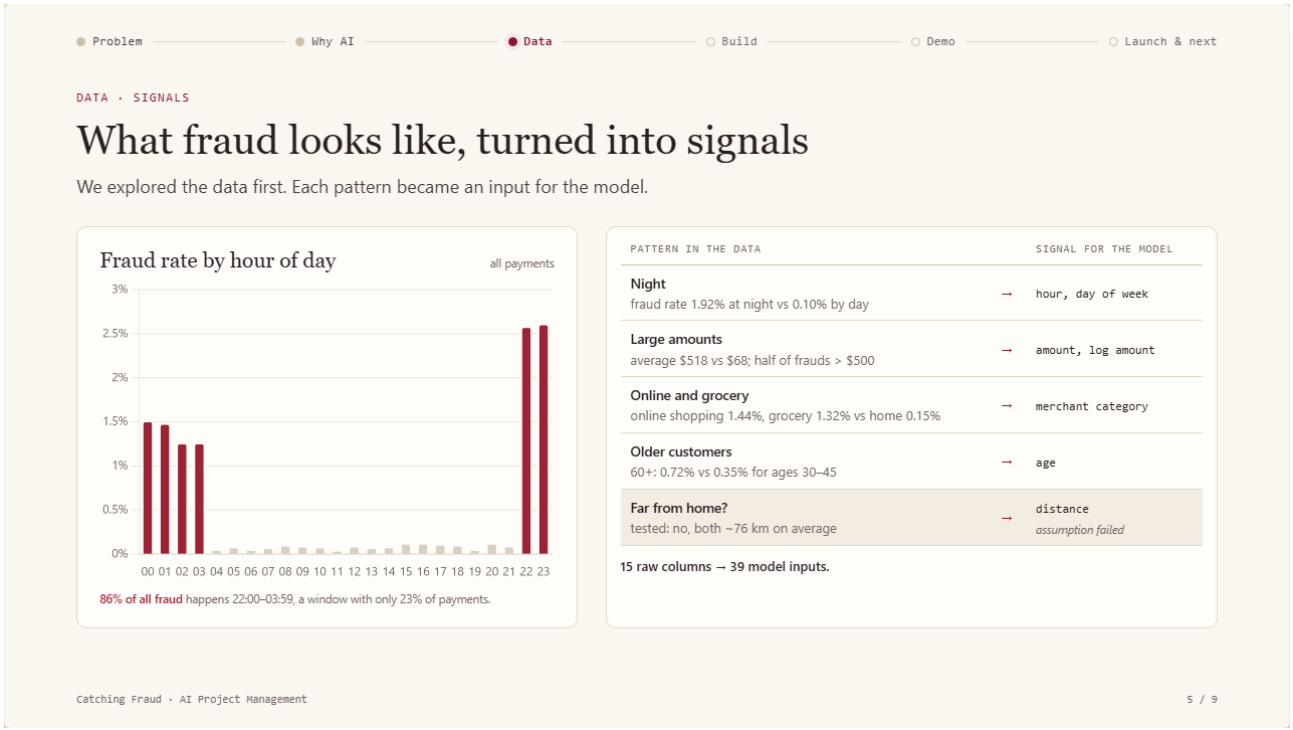


SPEAKER NOTES

- (~50 s) Our data: a public Kaggle dataset, 339,607 card payments from 13 western US states, 2019–2020. Lecture 2: an existing dataset is cheap and ready today, but it was collected by someone else for someone else.
- The core challenge: only 1,782 frauds, 1 in 191. A model hardly sees any fraud to learn from.
- Our fix: generated data. SMOTE creates new fraud examples by blending similar real frauds, until the training data is 50/50: 268,833 synthetic frauds made from 1,426 real ones. We also tested the alternative, class weights, where rare fraud simply counts more.
- The test set (67,922 payments) stays 100% real and untouched, so every result we show is on real payments.
- For a real bank: retrain on its own payments; the fraud label then comes from chargebacks weeks later. Check whether the Kaggle data is simulated, and say so if asked.

SLIDE 5 / 9 · DATA · SIGNALS

What fraud looks like

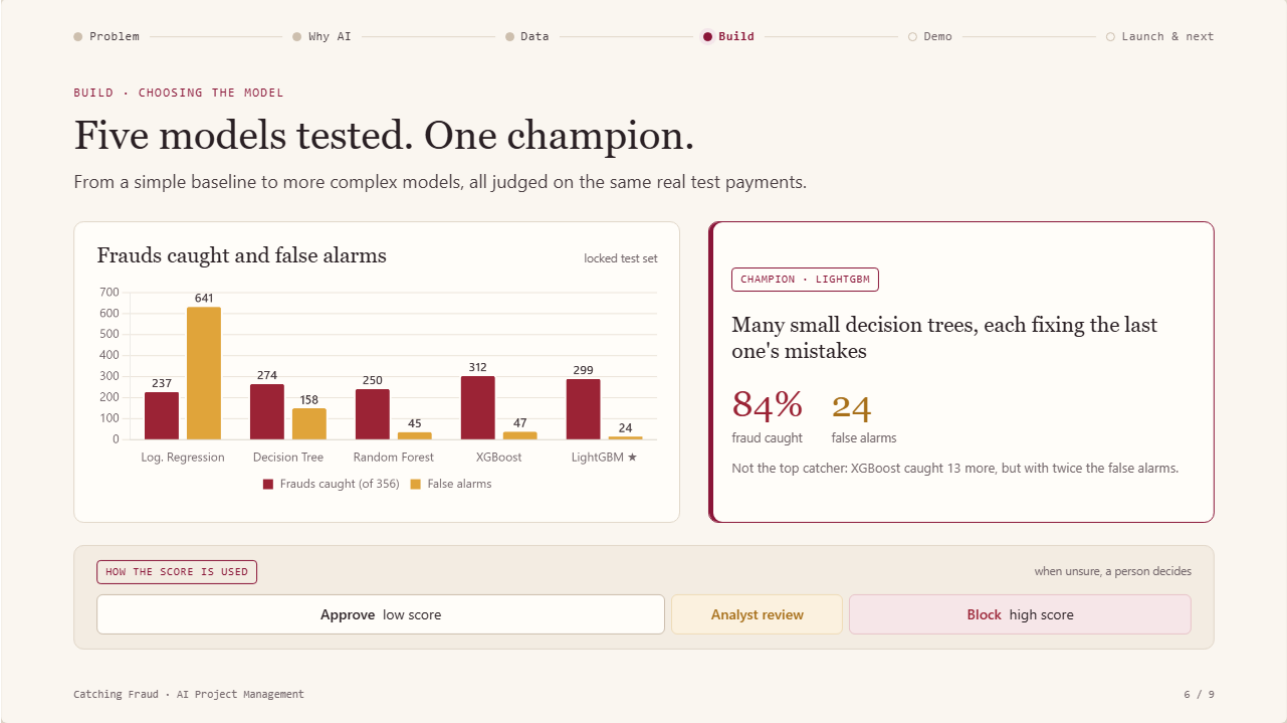


SPEAKER NOTES

- (~45 s) Before building any model, we looked at what makes fraud different, and turned each finding into a signal.
- Night: 86% of fraud happens between 22:00 and 04:00, a window with only 23% of payments. Amount: fraud averages $518 vs $68. Category: online shopping and grocery have the highest fraud rates. Age: customers over 60 are targeted twice as often as 30–45 year olds.
- We also tested an assumption: "fraud happens far from home". The data said no: both are about 76 km. Lesson: test assumptions with data.

SLIDE 6 / 9 · BUILD · CHOOSING THE MODEL

How we built it

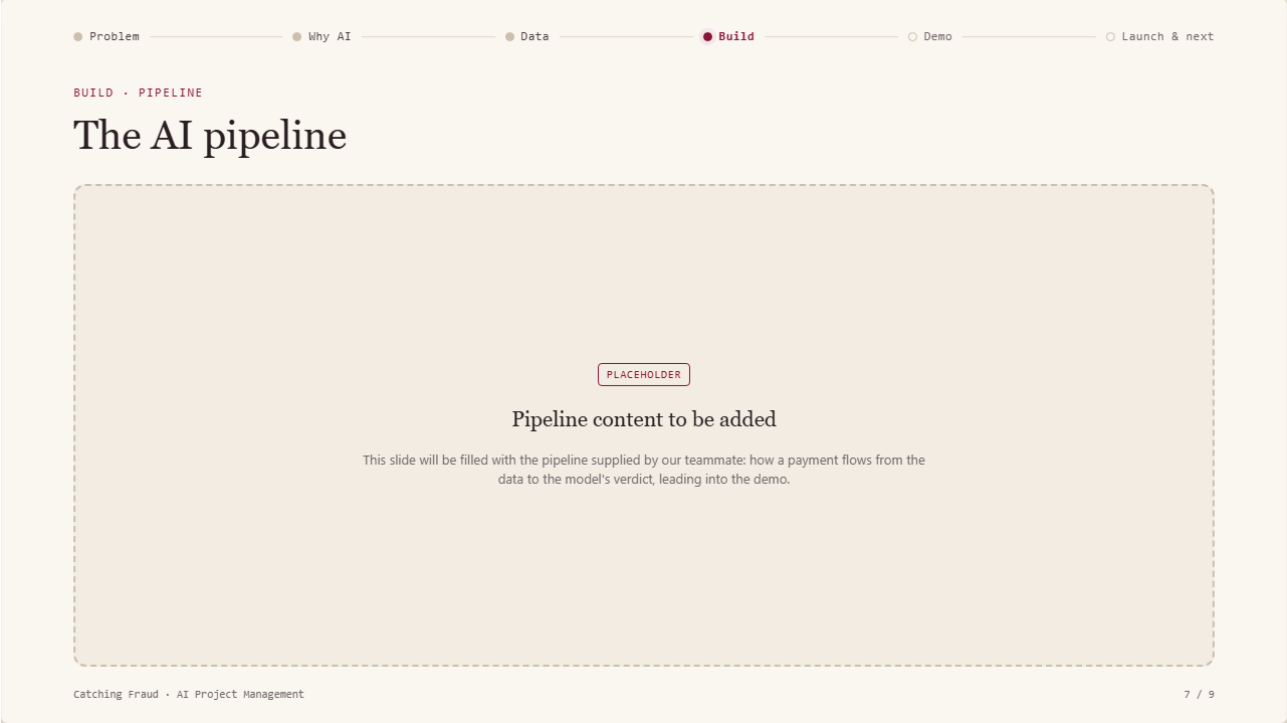


SPEAKER NOTES

- (~50 s) Lecture 3: start simple, change one idea at a time. We trained five models, from a simple straight-line model to tree ensembles, tuned each with cross-validation and judged all on the same locked test set.
- Baseline (Logistic Regression) caught 237 frauds with 641 false alarms. The champion, LightGBM, caught 299 (84%) with only 24 false alarms.
- We chose on balance, not on the single highest number: XGBoost caught 13 more but with 47 false alarms. Honest note: with only 356 frauds, a gap of 13 is within the noise (about ±14), so it's a business choice to confirm on live data.
- The model outputs a score. Low: approve. High (0.4 and up): block. In between (0.1–0.4): a fraud analyst decides. On our test set that middle band holds only about 23 payments, 6 of them fraud.
- Details in the backup deck: experiment log, threshold tuning, and the lesson that synthetic data made validation look too good.

SLIDE 7 / 9 · BUILD · PIPELINE

AI pipeline

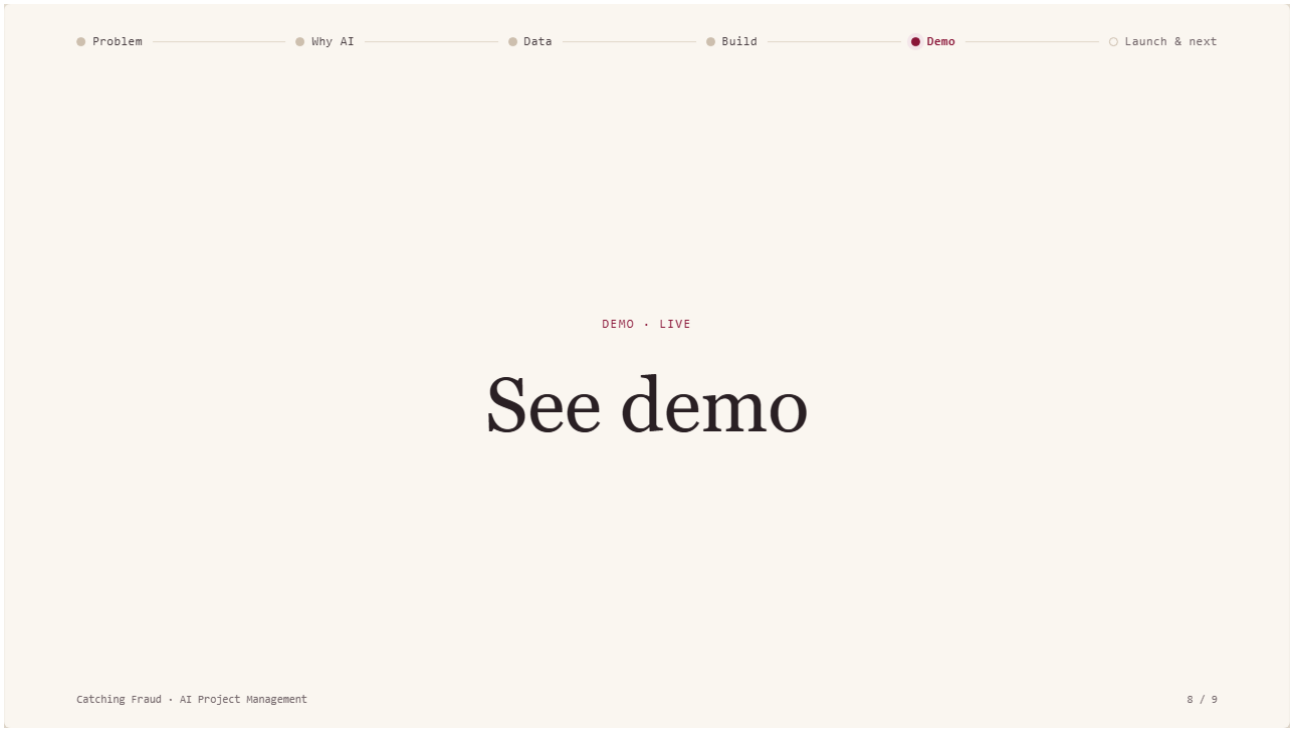


SPEAKER NOTES

- (~45 s) Placeholder: pipeline content and speaker notes to be provided by teammate.
- End by handing over to the demo: "Let's see it in action."

SLIDE 8 / 9 · DEMO · LIVE

Live demo

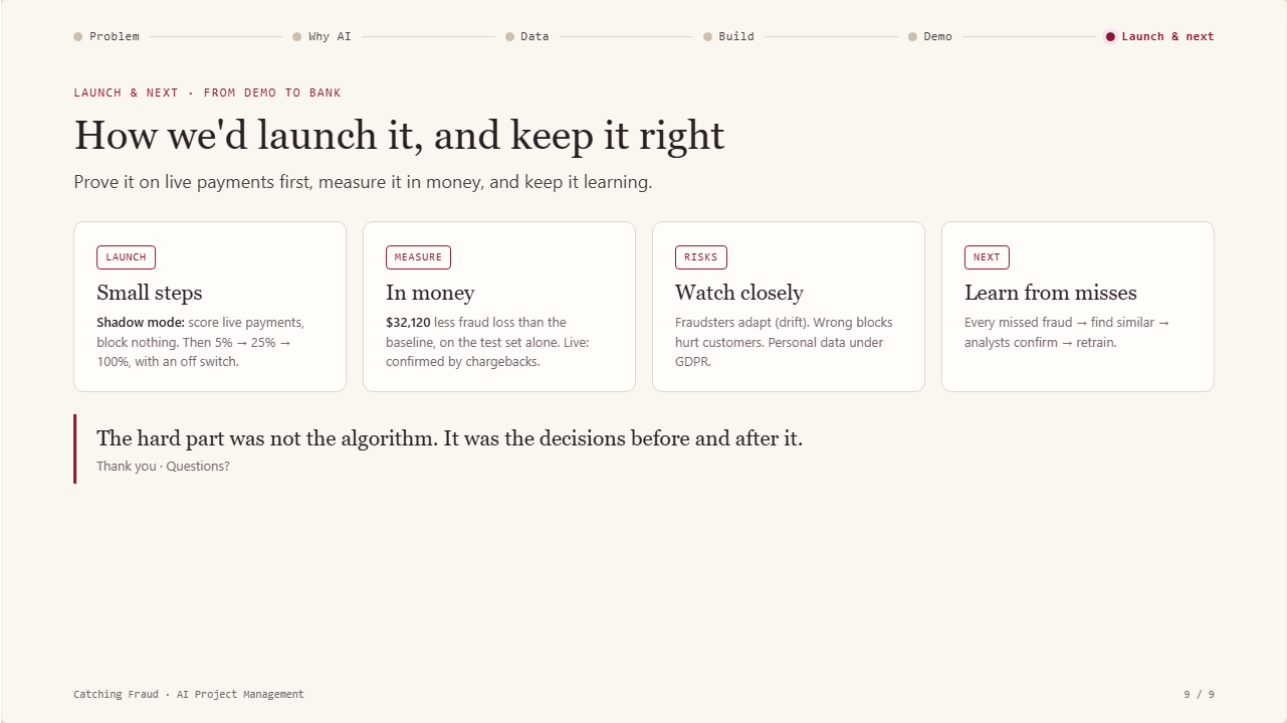


SPEAKER NOTES

- (~1:30) Switch to the demo window (Flask API on port 5050, Next.js dashboard on port 3000).
- Suggested order: an everyday purchase (approved), a payment with the fraud fingerprint: night, large amount, online (blocked), a borderline case (show how the score decides).
- Backup plan: have screenshots ready in case the demo fails.

SLIDE 9 / 9 · LAUNCH & NEXT · FROM DEMO TO BANK

Launch and next steps



SPEAKER NOTES

- (~50 s) Launch (Lecture 8): first shadow mode, where the model scores real payments but blocks nothing. Fraud is confirmed by chargebacks up to ~60 days later, so we wait before judging. Then a canary release: 5%, 25%, 100% of payments, always with a way back.
- Measure: success in money. On the test set, the champion prevents about $32,120 more fraud loss than the baseline model (62 more frauds × $518). Live, we'd also track false alarms, complaints and results per customer group.
- Risks: fraudsters change tactics, so the model ages; real customers wrongly blocked; personal data (birth date, location) under GDPR.
- Next (Lecture 2, data engine): each missed fraud becomes training data for the next model.
- Close with the one-liner, then open for questions. The backup deck has all details.